

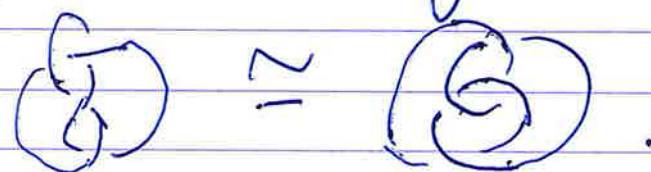
§1 Knot Invariants.

①

Defn
A knot is a piecewise linear emb. $S^1 \rightarrow S^3$.
w/ finitely many segments.

Theorems by people like Reidemeister show it's
OK to replace "p.l." with eq. "Smooth".

Natural eq. relⁿ: ambient isotopy in S^3 .

eg. 

Qn. How to tell two knots apart?
How to tell if a diagram represents the unknot?

Need to define invariants: computable funct^s

$$i: \{\text{diagrams}\} \rightarrow \{\text{some nice set}\}$$

eg. $\mathbb{Z}$, polyn.

such that if $k \stackrel{\text{Isot.}}{\sim} l$, $i(k) = i(l)$.

Big industry eg. ^{colours, numbers, "Doe type", hyp. volume,} Alexander polyn., Jones poly,
HOMFLY-PT, ...

Want invariants that are computable from
a diagram.

Knots are comp. det. by homo type.
Gordon & Luecke, 1989.
(for knots).
Not for links.

②

R. Riley '71: One family of invariants comes from reps

$$\rho: \pi_1(S^3 - K) \rightarrow \mathrm{PSL}(2, \mathbb{F}_p)$$

and their related homologies.

eg. reps into $\mathrm{PSL}(2, \mathbb{F}_7)$ distinguish

Kinoshita-
Terasaka
'57



and



Conway
'70.

which are indist. by Jones polynomials.

Qn. What about arithmetic reps

$$\rho: \pi_1(S^3 - K) \rightarrow \mathrm{PSL}(2, \mathbb{C})?$$

§ 2. $PSL(2, \mathbb{C})$.

(3)

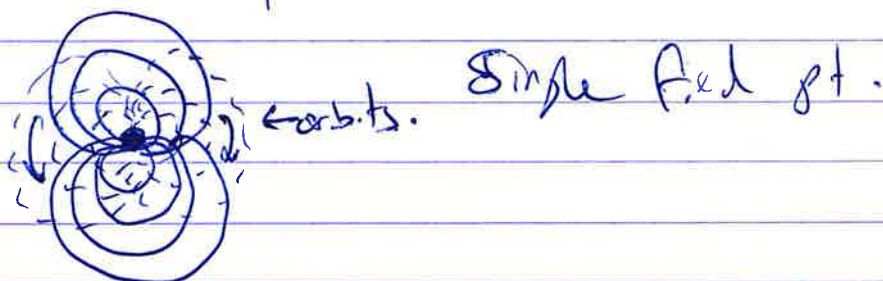
If $M = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \in PSL(2, \mathbb{C})$, $\det M \neq 1$,

then M acts on $\hat{\mathbb{C}} = \mathbb{C} \cup \{\infty\} = \mathbb{P}^1 \mathbb{C} = S^2$

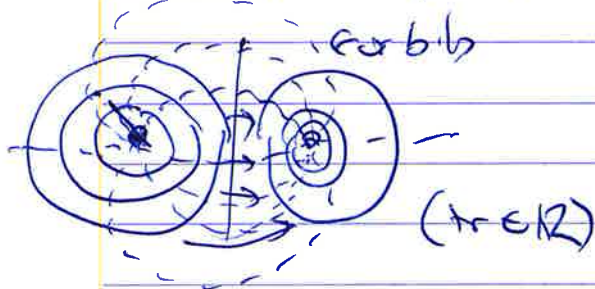
by $M \cdot z = \frac{\alpha z + \beta}{\gamma z + \delta}$.

3 cases:

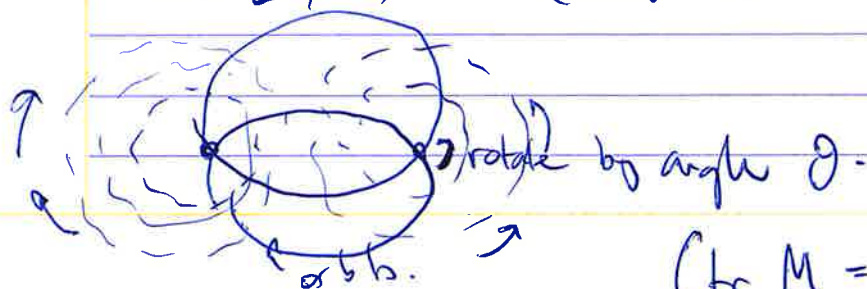
- $\text{tr}^2 M = 4$ (parabolic):



- $\text{tr}^2 M \notin [0, 4]$ (loxodromic):



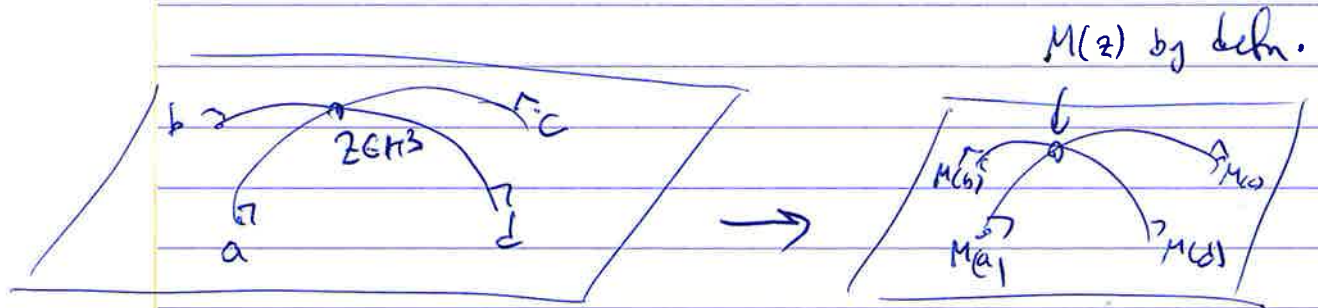
- $\text{tr}^2 M \in [0, 4)$ (elliptic):



$(\text{tr} M = 2 \cos \theta/2)$.

④

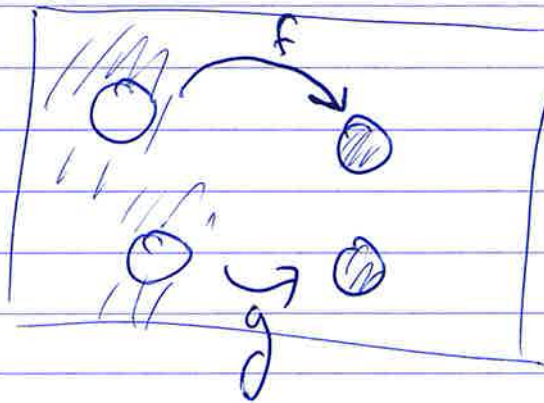
Poincaré observed that M also acts on $\mathbb{H}^3$:



This action is by isometries.

$\Rightarrow$ if $\Gamma < \text{PSL}(2, \mathbb{C})$ is discrete,
then $\mathbb{H}^3/\Gamma$ is a hyperbolic orbifold.

eg.



Thm. (Schottky, Fuchs-Klein)

if 2 pairs of d.f.s. Circles are given, and f, g send each of one into the int. of the other in pairs,
then $\langle f, g \rangle$ is discrete & free.

Also, $\mathbb{H}^3/\Gamma \cong \text{torus}$

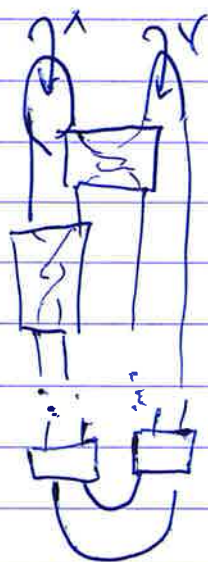
§3 The Riley slice.

⑤

Remember, Riley studied rep.

$$\pi_1(S^3 \setminus k) \rightarrow \mathrm{PSL}(2, \mathbb{C}).$$

Consider ~~each~~ the 2-bridge link:



$$\pi_1(S^3 \setminus k) = \langle X, Y : W=1 \rangle.$$

Thm. (Riley): $\exists$ a discrete faithful rep. $\pi_1(S^3 \setminus k) \rightarrow \mathrm{PSL}(2, \mathbb{C})$ iff k does not embed on a torus.

Cor. ('74): $\exists$ a knot k such that $S^3 \setminus k$ admits a complete hyp. metric

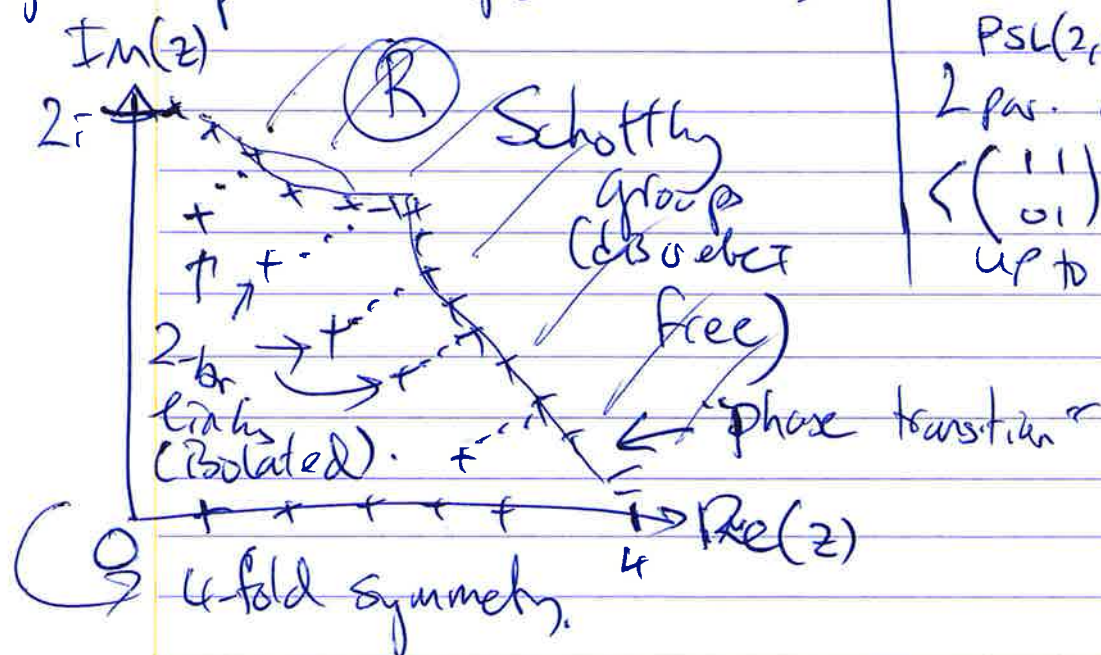
(major ~~not~~ influence on work of W. Thurston in late '70s).

~~Picture of all discrete reps of~~ ⑥

For hyp. geom. reasons, we consider only
 reps $\langle X, Y: W \rangle \rightarrow \mathrm{PSL}(2, \mathbb{C})$

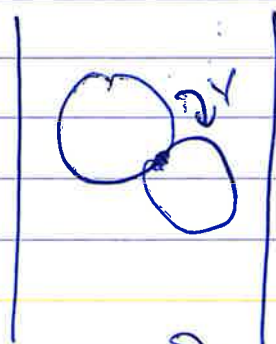
Such that X, Y are both parabolic.

Riley's Computer experiment ('71):



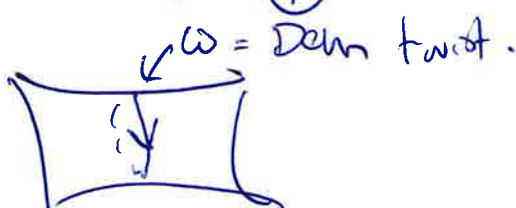
All subgroups of $\mathrm{PSL}(2, \mathbb{C})$ gen by
 2 par. are
 $\langle \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ z & 1 \end{pmatrix} \rangle$
 up to conj.

General group in the region ② has
 fund. dom. of the form



They are fundamental domains of
 hyperbolic str. on
 genus 2 handlebodies.

Thm (e.g. Ahlfors-Bers): ③ = $\mathrm{Teich}(S_{0,4}) / \langle W \rangle$



⑦

Boundary of $\mathbb{R}$ is highly fractal & hard to understand.

- Giving bounds on $\mathbb{R}$ is very important, and has applications to:

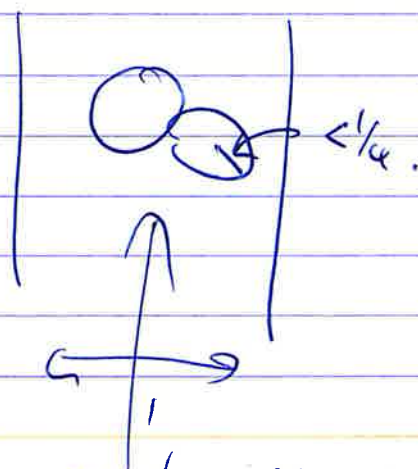
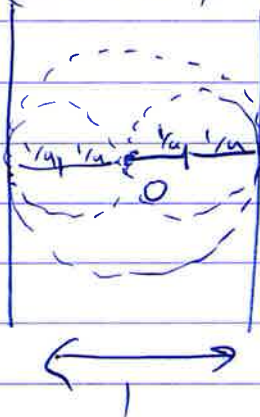
- enumeration of arithmetic groups in $PSL(2, \mathbb{R})$
- computation of ~~max~~ certain geometric minima (e.g. minimal volume manifolds).

~~Fractal~~ Fractal bounds:

- Bremer ('55): When $|z| > 4$,

$$\Gamma_z = \left\langle \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ z & 1 \end{pmatrix} \right\rangle \text{ is discrete \& free.}$$

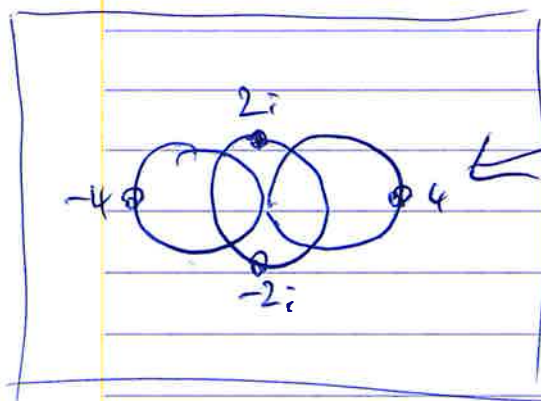
Proof: if $|z| > 4$ then the circles with centres $\pm \frac{1}{z}$ and radii $\frac{1}{|z|}$ define a fundamental domain for Γ_z :
(extremal:)



Circles lie inside the strip.

⑧

Chang - Serfling - Ree ('58):

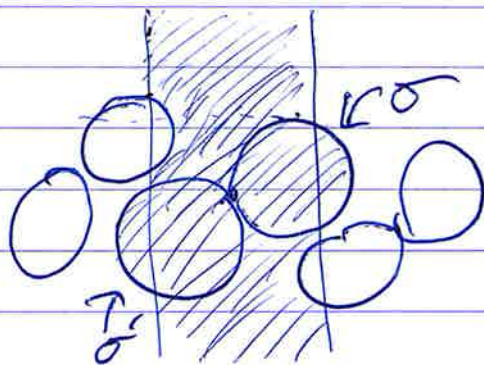


3 inequalities from the conditions for

$$X^{\pm 1} \left(B^{\pm \frac{1}{2}}, \frac{1}{|z|} \right)$$

are all disjoint:

Proof:



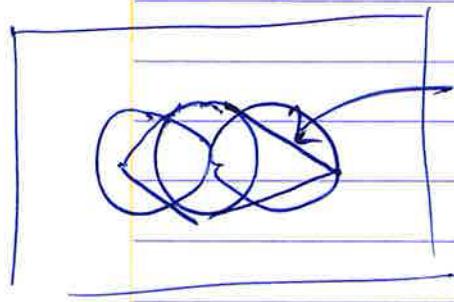
X sends shaded region  into unshaded region.

Y sends  into  since it maps exterior of σ into interior of σ^* .

Ping-pong lemma: disjoint & free.

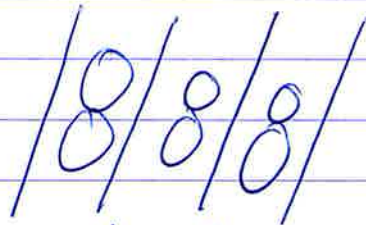
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Lyndon - Ullman (1999):



improved to
convex hull of
 $\{ \pm 4, B(0, 2) \}$.

Method: modification of pg-pg sub
to



skewed strips.

§ 4. Elliptic slices.

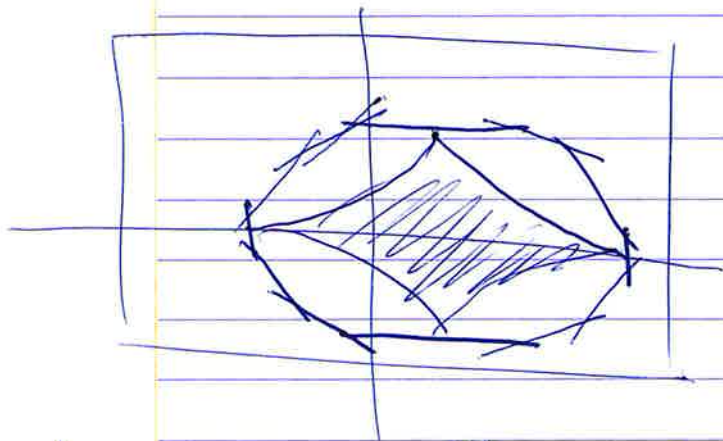
Defn. $R^{p,q} = \{ z : \begin{pmatrix} e^{i\pi/p} & 1 \\ 0 & e^{-i\pi/p} \end{pmatrix}, \begin{pmatrix} e^{i\pi/q} & 0 \\ z & e^{-i\pi/q} \end{pmatrix} \}$
 B discrete & non,
 $\rightarrow \mathbb{Z}/p\mathbb{Z} * \mathbb{Z}/q\mathbb{Z}$.

Example of $PSL(2, \mathbb{C})$ gen. by two
elliptic matrices of order p, q & conjugate
to some Γ_z .

(10)

Thm. (E. - Gough - Martin - Schlicht, '24).
(to appear Ann. Inst. Fourier, '26).

$\mathbb{C} \setminus \mathbb{R}^{p,q}$ P contained within a dodecagon:



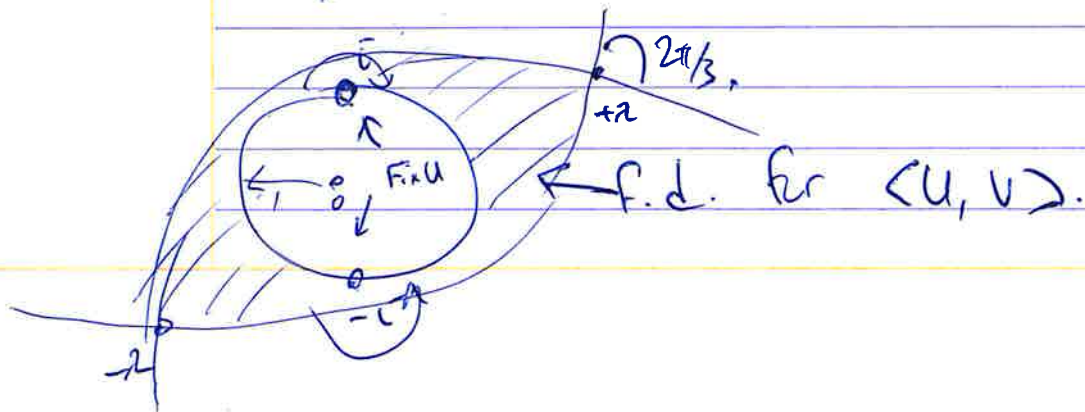
A simple version. Fix $p=2, q=3$; this
is motivated by studying discrete reps. of
 $PSL(2, \mathbb{C}) \cong \mathbb{Z}/2\mathbb{Z} * \mathbb{Z}/3\mathbb{Z} \rightarrow PSL(2, \mathbb{C})$.

One choice of gens:

$$\Gamma_2 = \langle U = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, V = \begin{pmatrix} 1/2 & \sqrt{3}/2 \\ \sqrt{3}/2 & 1/2 \end{pmatrix} \rangle.$$

lem. Γ_2 is discrete & $\cong \langle U \rangle * \langle V \rangle$

if the follo~~w~~ing circles are disjoint:



(11)

This is equiv. to.

$$|2 \cot \frac{\pi}{3} \pm \cot \frac{\pi}{2}| + \csc \frac{\pi}{2} \leq |2| \csc \frac{\pi}{3}$$

i.e. $\frac{|2|}{\sqrt{3}} + 1 \leq \frac{2}{\sqrt{3}} |2| \Leftrightarrow \sqrt{3} \leq |2|.$

~~$$\sqrt{3} \leq 2|2|.$$~~

Another way of writing the generic repⁿ of $PSL(2, \mathbb{Z}) \simeq \mathbb{Z}/2\mathbb{Z} * \mathbb{Z}/3\mathbb{Z}$:

$$G_t = \left\langle \frac{1}{\sqrt{-t}} \begin{pmatrix} -t & 1 \\ 0 & 1 \end{pmatrix}, \frac{1}{\sqrt{-t}} \begin{pmatrix} 1 & 0 \\ t & -t \end{pmatrix} \right\rangle.$$

This is essentially the image of B_3 the Burau repⁿ of B_3 in $PSL(2, \mathbb{C})$.

Change of var $t \leftrightarrow \lambda$:

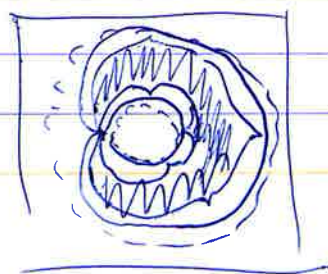
$$\lambda = \frac{\sqrt{3}(\mu \pm \sqrt{\mu^2 + \mu + 1}) - 1}{3\sqrt{\mu}}.$$

Pushing $|2| \geq \sqrt{3}$ through, we find that,

if $z = \sqrt{t} - \frac{1}{\sqrt{t}}$, and if

$$3 \leq |z \pm \sqrt{z^2 + 3}|,$$

$$\text{then } G_t \simeq B_3 / \mathbb{Z}(B_3) \cong \mathbb{A}.$$



Since $G_t = PSL(2, \mathbb{Z})$ for $q = -t$, we prove:

Conj. (Morier-Genoud, Ossereto, Veselov): If $\frac{3-\sqrt{5}}{2} |q| \leq \frac{3+\sqrt{5}}{2}$, then $PSL(2, \mathbb{Z}) \cap \Gamma_q \neq \{1\}$.